

PUFormer: A Physics-Unfolded Transformer With Exact Observation Operators for Hyperspectral and Multispectral Image Fusion

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Abstract—Fusing a low-resolution hyperspectral image (LR-HSI) with a high-resolution multispectral image (HR-MSI) is an ill-posed inverse problem whose forward model is usually known exactly in benchmark settings, yet most deep fusion networks either ignore the model or apply it only approximately. We present PUFormer, a physics-unfolded transformer that alternates exact data-consistency steps with a learned spectral-attention prior. Each stage takes a gradient step on both observation residuals using the true blur-decimation operator and spectral response and their exact adjoints, including the reflect-padded image border that the data simulation uses, and then corrects the result with a three-level U-Net of transposed-attention transformer blocks that carries features from one stage to the next. The network is trained with an L1 loss, deep supervision and small spectral-angle and structural-similarity terms. On the Chikusei $\times 4$ benchmark with a WorldView-2 multispectral sensor, under the protocol of a recent IEEE TIP study that compared 17 methods, PUFormer reaches 58.20 dB PSNR, a spectral angle of 0.713° , an ERGAS of 1.330 and an RMSE of 18.9 digital numbers with 10.1 M parameters, ranking first on seven of the eight reported metrics. On Pavia Centre $\times 4$ with a four-band IKONOS-type sensor, a model trained from scratch in one 12-hour session reaches 50.69 dB, 3.53 dB above the best reported value, with a 34% lower RMSE, and ranks first on six of the eight metrics. The reconstructions reproduce both inputs to within 77–90 dB, and when the true blur is supplied at test time the models stay within 0.26 dB of their nominal accuracy for blur widths they were not trained on. We further show that the remaining gaps, Q2n on Chikusei and SAM and Q2n on Pavia Centre, come from spectra that the inputs cannot determine: ultraviolet bands outside the multispectral sensor’s range, and dark, noise-dominated water pixels. We release the code, logs and evaluation scripts.

Index Terms—Hyperspectral image fusion, multispectral image, deep unfolding, transformer, super-resolution, Chikusei, Pavia Centre.

I. INTRODUCTION

HYPERSPECTRAL imagers record hundreds of narrow spectral bands but, for a fixed photon budget, at a coarse spatial resolution; multispectral imagers do the opposite. Hyperspectral and multispectral image fusion recovers a high-resolution hyperspectral image (HR-HSI) from one observation of each kind [1]–[3]. Classical methods formulate fusion as a regularized inverse problem, using component substitution [4], coupled matrix factorization [5], subspace regularization [6] or fast Sylvester-equation solvers [7]. Deep networks have

since become the dominant approach, from convolutional reconstruction networks [8], [9] to transformers [10]–[13] and diffusion models [14]–[17].

Two observations motivate this work. First, in the simulated benchmarks on which fusion methods are ranked, the degradation from the HR-HSI to each observation is known exactly, yet most networks treat both observations as generic inputs. Model-guided and unfolded networks [18]–[22] do use the operators, but they typically apply zero-padded or approximate versions that do not match the operator that generated the data at the image border, so even the ground truth fails the network’s own consistency check there. Second, reported numbers on the same dataset vary by tens of decibels across papers because of differences in the sensor response, blur, split, normalization and even the definition of each metric [3]. Claims of improvement are therefore only meaningful within one fully specified protocol.

We address both points on two datasets under the protocols of Chen *et al.* [15], which report 17 methods on eight metrics for each: Chikusei [23] with a WorldView-2 multispectral response and eight 256×256 test images, and Pavia Centre [24] with an IKONOS response and four 256×256 test images, both with a 7×7 Gaussian blur ($\sigma = 2$) and $\times 4$ decimation. Our contributions are:

- 1) **PUFormer**, a physics-unfolded network whose data-consistency steps use the exact forward operators and their exact adjoints, including the reflect-padded border of the simulation, combined with a transposed-attention U-Net prior [25] that carries memory across stages.
- 2) A **complete and reproducible evaluation**: eight metrics implemented from their reference code, three PSNR and two SSIM definitions reported side by side, per-image results, and the consistency of each reconstruction with its own inputs.
- 3) **State-of-the-art accuracy** on both protocols with about 10 M parameters: 58.20 dB PSNR on Chikusei (+2.01 dB over the best reported method, first on seven of eight metrics) and 50.69 dB on Pavia Centre (+3.53 dB, first on six of eight).
- 4) **Robustness and failure analysis**: a test-time operator swap that almost fully compensates blur mismatch on both datasets, a noise study, and analyses showing that the Q2n shortfall on Chikusei comes from bands without multispectral coverage and the SAM and Q2n shortfalls on Pavia Centre from one test image with water.

E-mail: amarnath10chinu@gmail.com. Code, training logs and all result files are available at <https://github.com/Amarnath10i/Research-On-Hyperspectral-ImageFusion>.

II. RELATED WORK

A. Model-based fusion

Component substitution methods such as GSA [4] inject multispectral detail into the upsampled hyperspectral bands. Unmixing-based methods such as CNMF [5] factor both observations with shared endmembers, and subspace methods such as HySure [6] regularize the HR-HSI within a low-dimensional spectral subspace, whose dimension can be estimated with HySime [26]. FUSE [7] solves the resulting quadratic problem in closed form through a Sylvester equation. These methods use the forward model explicitly but rely on hand-crafted priors.

B. Supervised deep fusion

SSR-NET [8] and GuidedNet [9] learn a direct mapping from the two observations. Transformers model long-range spatial and spectral dependencies: PSRT [10] uses pyramid shuffle-and-reshuffle attention, MIMO-SST [11] multi-input multi-output supervision, LGCT [12] local-global collaboration, DSPNet [27] a dual spatial-spectral pyramid, SSDT [13] dilated attention, and MIMFormer [28], AMSF-Net [29], ASMNet [30] and DPFormer [31] multiscale spectral-spatial designs. Recent work also studies state-space models [32], [33], mixture-of-experts routing [34] and modality interaction [35]. General image-fusion frameworks such as U2Net [36] and OTPNet [37] have also been applied to this task.

C. Model-guided and unfolded networks

Algorithm unrolling [38]–[40] turns the iterations of an optimization method into network stages. MHF-Net [18] and MoG-DCN [19] unfold fusion-specific solvers, DUNET [20] adds transformer priors for unregistered inputs, HMPNet [21] extends unfolding to three modalities, and an information-aware transformer-based unfolding network [22] and a mutual-guidance unfolding network (SMGU) [41] strengthen the learned proximal steps. PUFormer belongs to this family. It differs in using the exact simulation operator at the image border, in its cross-stage memory and in its transposed-attention prior, which attends across spectral feature channels rather than spatial windows.

D. Generative fusion

Diffusion models have been conditioned on the two observations (DDIF [14]), constrained by the observation model during sampling (CDPS [16]), restricted to a learned subspace (ARGS-Diff [17]) or combined with differential attention in a two-stage conditioning scheme [15]. Null-space formulations [42] make the data-consistency part of the reconstruction exact. The two-stage conditional diffusion model of Chen *et al.* [15] holds the best published Chikusei $\times 4$ results in our survey and is our main reference; its comparison also includes the correlation-aware network CLSNet [43] and the dual cross transformer (DCTransformer) [44], denoted DCT in their table.

III. OBSERVATION MODEL AND EXACT OPERATORS

Let $\mathbf{X} \in \mathbb{R}^{B \times H \times W}$ be the HR-HSI with B bands ($B = 128$ for Chikusei and $B = 92$ for Pavia Centre). The two observations are

$$\mathbf{Y}_H = \mathcal{D}\mathbf{X}, \quad \mathbf{Y}_M = \mathcal{R}\mathbf{X}, \quad (1)$$

where $\mathcal{D} = \mathcal{S}\mathcal{K}\mathcal{P}$ applies, band by band, reflect padding $\mathcal{P}$ of width $p = 3$, a valid correlation $\mathcal{K}$ with a normalized 7×7 Gaussian kernel ($\sigma = 2$), and decimation $\mathcal{S}$ that keeps every fourth pixel starting at index zero, so $\mathbf{Y}_H \in \mathbb{R}^{B \times H/4 \times W/4}$. The spectral operator $\mathcal{R} \in \mathbb{R}^{M \times B}$ integrates the spectrum with the $M = 8$ WorldView-2 band responses [45] for Chikusei and with the $M = 4$ IKONOS band responses [46] for Pavia Centre. Because the manufacturers publish the responses only as plots together with band edges, we model each band as a flat top with logistic flanks (4 nm roll-off) that pass exactly through the published edges, at 5% response for WorldView-2, whose edges are tabulated at that level, and at 50% for the IKONOS band limits (blue 445–516, green 506–595, red 632–698 and near infrared 757–853 nm), and normalize every band to unit sum.

Exact adjoint. Every gradient step on $\|\mathcal{D}\mathbf{X} - \mathbf{Y}_H\|^2$ requires $\mathcal{D}^\top = \mathcal{P}^\top \mathcal{K}^\top \mathcal{S}^\top$. $\mathcal{S}^\top$ inserts zeros and $\mathcal{K}^\top$ is a full convolution; together they are a strided transposed convolution with an output padding of three. The adjoint of reflect padding folds the padded margins back onto the pixels they were copied from. For a padded row index $j \in [0, H + 2p]$ the source row is $p - j$ for $j < p$, $j - p$ in the interior and $2H - 2 + p - j$ in the bottom margin, hence

$$(\mathcal{P}^\top \mathbf{g})_i = \mathbf{g}_{i+p} + [1 \leq i \leq p] \mathbf{g}_{p-i} + [H - 1 - p \leq i \leq H - 2] \mathbf{g}_{2H-2+p-i}, \quad (2)$$

applied to rows and then to columns. Most unfolded networks instead use zero padding inside the network while the data are generated with a different border rule; the ground truth then does not satisfy (1) in the network's own operator, and the data-consistency step pushes the first rows and columns of every image in the wrong direction. With (2), $\mathcal{D}$ in the network is exactly the simulation operator, and the inner-product test $\langle \mathcal{D}\mathbf{X}, \mathbf{Y} \rangle = \langle \mathbf{X}, \mathcal{D}^\top \mathbf{Y} \rangle$ holds to a relative error of 10^{-16} in double precision. $\mathcal{R}^\top$ is the transpose of the response matrix.

IV. PUFORMER

A. Unfolded iterations

PUFormer (Fig. 1) starts from the bicubic upsampling $\mathbf{X}_0$ of $\mathbf{Y}_H$ and performs $K = 3$ stages. Stage k first takes a gradient step on both data terms,

$$\begin{aligned} \mathbf{Z}_k = & \mathbf{X}_{k-1} - 16 \alpha_k \mathcal{D}^\top (\mathcal{D}\mathbf{X}_{k-1} - \mathbf{Y}_H) \\ & - \beta_k \mathcal{R}^\top (\mathcal{R}\mathbf{X}_{k-1} - \mathbf{Y}_M), \end{aligned} \quad (3)$$

with learnable step sizes $\alpha_k = \text{softplus}(a_k)$ and $\beta_k = \text{softplus}(b_k)$. The factor 16 compensates the $1/16$ gain of $\mathcal{D}^\top \mathcal{D}$ on smooth content under $\times 4$ decimation, so that both steps start at a comparable scale. The prior then predicts a residual,

$$\begin{aligned} (\Delta_k, \mathbf{F}_k) = & \mathcal{P}_k([\mathbf{Z}_k, \mathbf{Y}_M, 16 \mathcal{D}^\top (\mathcal{D}\mathbf{Z}_k - \mathbf{Y}_H)], \mathbf{F}_{k-1}), \\ \mathbf{X}_k = & \mathbf{Z}_k + \Delta_k, \end{aligned} \quad (4)$$

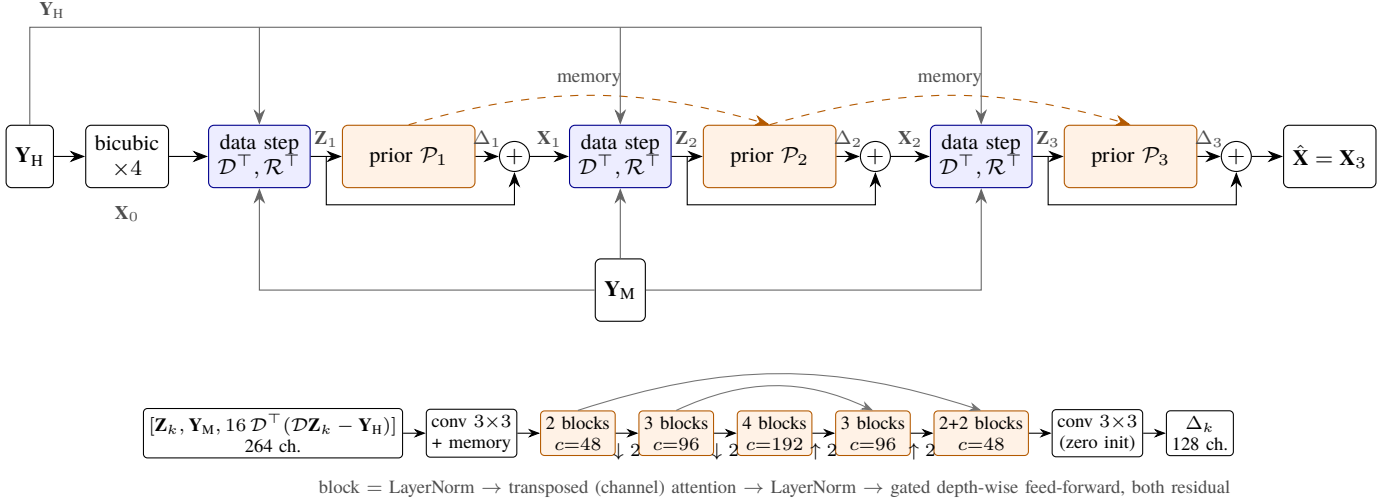


Fig. 1. PUFormer. Top: three unfolded stages, each an exact data-consistency step on both observations followed by a learned residual prior; decoder features are passed to the next stage’s prior (memory). Bottom: the prior of stage k , a three-level U-Net of transposed-attention transformer blocks (channel counts for Chikusei; for Pavia Centre the input has 188 and the output 92 channels).

where the third input tells the prior where the updated estimate still disagrees with $\mathbf{Y}_H$, and $\mathbf{F}_{k-1}$ are the final decoder features of the previous stage (cross-stage memory, merged by a 1×1 convolution). The last convolution of every prior is initialized to zero, so training starts from the pure physics iteration. The image path, including all operator applications, stays in 32-bit floating point; only the priors run in 16-bit mixed precision [47].

B. Transposed-attention prior

Each prior is a three-level U-Net [48] with widths 48, 96 and 192, two, three and four transformer blocks on the encoder levels and bottleneck, mirrored decoder levels, and two refinement blocks. A block follows Restormer [25]: multi-Dconv-head transposed attention computes queries, keys and values with 1×1 and depth-wise 3×3 convolutions, normalizes each feature map over space, and forms a $c' \times c'$ attention matrix across channels per head (1, 2 and 4 heads on the three levels) with a learned temperature. Attention is thus computed between spectral feature maps rather than between spatial positions, its cost is linear in the number of pixels, and the receptive field of every block covers the whole image. A gated depth-wise feed-forward network with expansion 2.66 follows. The three priors hold 3.37–3.38 M parameters each; the whole network has 10.13 M parameters for Chikusei and 9.99 M for Pavia Centre, whose input and output convolutions are smaller.

C. Training objective

With ground truth $\mathbf{X}$, the loss is

$$\mathcal{L} = \|\mathbf{X}_K - \mathbf{X}\|_1 + 0.1 \sum_{k < K} \|\mathbf{X}_k - \mathbf{X}\|_1 + \lambda_{\text{SAM}} \text{SAM}(\mathbf{X}_K, \mathbf{X}) + \lambda_S (1 - \text{SSIM}(\mathbf{X}_K, \mathbf{X})), \quad (5)$$

with mean absolute errors, deep supervision of the intermediate stages, the mean spectral angle in radians, and a band-wise Gaussian SSIM [49] whose data range is the maximum

Algorithm 1 PUFormer inference

Require: $\mathbf{Y}_H$, $\mathbf{Y}_M$, exact operators $\mathcal{D}$, $\mathcal{D}^\top$, $\mathcal{R}$, $\mathcal{R}^\top$

- 1: $\mathbf{X}_0 \leftarrow \text{bicubic}_{\times 4}(\mathbf{Y}_H)$, $\mathbf{F}_0 \leftarrow \emptyset$
- 2: **for** $k = 1, \dots, K$ **do**
- 3: $\mathbf{Z}_k \leftarrow \text{data step (3)}$ ▷ fp32
- 4: $\mathbf{r}_k \leftarrow 16 \mathcal{D}^\top(\mathcal{D}\mathbf{Z}_k - \mathbf{Y}_H)$
- 5: $(\Delta_k, \mathbf{F}_k) \leftarrow \mathcal{P}_k([\mathbf{Z}_k, \mathbf{Y}_M, \mathbf{r}_k], \mathbf{F}_{k-1})$ ▷ fp16
- 6: $\mathbf{X}_k \leftarrow \mathbf{Z}_k + \Delta_k$
- 7: **end for**
- 8: **return** $\mathbf{X}_K$

of each training patch. We use $\lambda_{\text{SAM}} = 0.05$ and $\lambda_S = 0.1$; both terms are computed in 32-bit precision. An exponential moving average of the weights (decay 0.999) [50] is used for validation and testing.

D. Self-ensemble

At test time we average the prediction for the input with the transposed prediction for the transposed input [51]. Transposition preserves the decimation phase, so both passes see observations generated by exactly the same operator. Flips and rotations do not: flipping a 256-pixel row maps sampled position $4m$ to $255 - 4m$, which has phase three, so we do not use them.

V. EXPERIMENTAL SETUP

A. Data and protocol

Chikusei [23] is an airborne scene of 2517×2335 pixels with 128 bands from 363 to 1018 nm. The split sizes follow [15]: eight 256×256 test images, 64 validation patches and 832 training patches of 64×64 . Because their locations are not specified there, we fix them as follows. We take the central 2048×2048 crop, which contains no no-data border, and divide it by its global maximum (15,133 digital

TABLE I
THE TWO $\times 4$ PROTOCOLS (DEGRADATION AND SPLIT SIZES OF [15]; IMAGE LOCATIONS OURS)

Item	Chikusei	Pavia Centre
HR-HSI	central 2048 ² crop, 128 bands	1096 $\times$ 715, bands 11–102 (92)
Normalization	$\div 15,133$ DN	$\div 8,000$ DN
LR-HSI	Gaussian 7×7 , $\sigma = 2$; $\times 4$ decimation; reflect border	
HR-MSI	WorldView-2, 8 bands	IKONOS, 4 bands
Test	8 of 256 ² (rows 0–256)	4 of 256 ² (rows, columns 0–512)
Validation	64 of 64 ² (rows 272–400)	63 of 64 ² (rows 528–656)
Training	random 64 ² crops, rows 416–2048	random 64 ² crops, see text
Selection	validation PSNR only	

numbers, DN). The test images are the eight 256×256 tiles of the top 256 rows, the validation patches tile rows 272–400, and training draws random 64×64 crops with dihedral augmentation from rows 416–2048 (800 non-overlapping patch equivalents). Sixteen-row gaps keep the three regions disjoint.

Pavia Centre [24] was recorded by the ROSIS sensor over the centre of Pavia, Italy; the distributed scene has 1096×715 pixels at 1.3 m and 102 bands between 430 and 860 nm. For this dataset, [15] uses the benchmark files released with PSRT [10]: 92 bands, a four-band multispectral image, four 256×256 test images and 142 validation and 1284 training patches of 64×64 . These files are distributed through a file-sharing service that we could not access, so we rebuild the benchmark from the original scene to the same specification. We keep bands 11–102 (473–860 nm if the 102 bands are taken as evenly spaced over 430–860 nm), divide by the maximum of the retained bands (8,000 DN), apply the same blur and decimation as for Chikusei, and simulate the HR-MSI with the IKONOS responses. The test images are the four 256×256 tiles of the top-left 512×512 region. The 63 validation patches lie on a 32-pixel grid in rows 528–656, and training draws random 64×64 crops with dihedral augmentation from rows 672–1096 and from the strip to the right of the test region (rows 0–512, columns 528–715), 97 non-overlapping patch equivalents in total. Gaps of 16 pixels keep the regions disjoint. As a check on the rebuilt data, our GSA implementation reaches 34.78 dB on our test images, against 35.35 dB reported for GSA in [15].

For both datasets we follow Wald’s protocol [52]: the original image serves as ground truth and both observations are simulated from it, and every image, including every training crop, is degraded on its own with (1). Table I summarizes the two protocols.

B. Metrics

We report the eight metrics of [15], averaged over the test images: PSNR; SSIM [49]; the spectral angle mapper (SAM, degrees) [53]; ERGAS with ratio 4 [54]; Q2n [55] on 32×32 blocks, computed on the image in DN exactly as in the reference implementation of the pansharpening toolbox [56], [57]; the band-averaged correlation coefficient (CC); the spatial correlation coefficient (SCC) [58] of Sobel gradient magnitudes as in the same toolbox; and RMSE in DN. The headline PSNR is $10 \log_{10}(1/\text{MSE})$ over the whole normalized image, i.e., its peak is the dataset maximum. The PSNR values of [15] also use a fixed peak: for every row

TABLE II
TRAINING RUNS

	Chikusei		Pavia Centre
	First run	v3 (final)	final
Initialization	random	first run	random
Border of $\mathcal{D}$	zero padding	reflect (exact)	reflect (exact)
Loss	L1 + 0.1 deep sup.	(5)	(5)
GPUs $\times$ batch	1×8	2×8	2×8
Peak learning rate	3×10^{-4}	1.5×10^{-4}	3×10^{-4}
Iterations	88,946	94,450	93,850
Training patches	0.71 M	1.51 M	1.50 M
Wall-clock time	9.5 h	10.9 h	10.2 h
Selected at	best val. (58.60 dB)	it. 44,000 (58.72 dB)	it. 43,000 (48.58 dB)

of their Chikusei table, $\text{RMSE} \cdot 10^{\text{PSNR}/20}$ is between 16.6k and 17.1k DN. Since our peak is lower (15,133 DN), our headline PSNR is conservative; with a 16.7k DN peak it would be 0.86 dB higher. For Pavia Centre the same product lies between 7.7k and 8.4k DN (median 8.07k), close to our peak of 8,000 DN, so there the PSNR values are on the same scale. The SSIM definition of [15] is not stated. We report SSIM as in the code released with PSRT [10], whose benchmark data [15] adopts (Gaussian 11×11 window, $\sigma = 1.5$, data range 1, i.e., the dataset peak), and additionally a stricter variant whose data range is each image’s own maximum. We also report observation consistency, $\text{PSNR}(\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_H)$ and $\text{PSNR}(\mathcal{R}\hat{\mathbf{X}}, \mathbf{Y}_M)$.

C. Implementation

PUFormer is implemented in PyTorch [59] and trained on Kaggle NVIDIA Tesla T4 GPUs (Table II). On Chikusei we train in two runs. The first run trains from scratch on one GPU with zero-padded operators and the L1 losses only. The final model (v3) fine-tunes it with the exact reflect-border operators and the full loss (5) on two GPUs with distributed data parallelism [60]. On Pavia Centre, a single run trains the model from scratch with the exact operators and the full loss on two GPUs, within one 12-hour session that also runs all tests. All runs use AdamW [61] ($\beta = (0.9, 0.99)$, weight decay 10^{-4}), a 1000-iteration linear warm-up followed by a cosine decay [62] to 10^{-3} of the peak rate that ends at the time budget, and gradient-norm clipping at 0.1. The weights with the best validation PSNR are tested once.

VI. RESULTS

A. Comparison with the state of the art on Chikusei

Table III lists all Chikusei results of [15] together with PUFormer. The compared methods are GSA [4], CNMF [5], FUSE [7], U2Net [36], PSRT [10], DCT [44], DDIF [14], MIMFormer [28], MIMO-SST [11], OTPNet [37], SMGU [41], AMSF [29], SSDT [13], ASMNet [30], DPFormer [31], CLSNet [43] and the two-stage conditional diffusion model [15]; their values are copied from [15].

PUFormer ranks first of 18 on PSNR, SSIM (PSRT definition), SAM, ERGAS, CC, SCC and RMSE, with or without the self-ensemble. Relative to the best reported method, PSNR improves by 2.01 dB, RMSE by 26.4%, ERGAS by 12.2% and SAM by 1.1%. It ranks tenth on Q2n, which Section VI-F

TABLE III

CHIKUSEI $\times 4$ COMPARISON. ALL ROWS EXCEPT PUFORMER ARE REPORTED IN [15]. BEST VALUE PER COLUMN IN BOLD. SSIM OF PUFORMER USES THE PSRT DEFINITION (DATA RANGE 1); WITH DATA RANGE EQUAL TO EACH IMAGE'S MAXIMUM IT IS 0.9969 (SINGLE PASS) AND 0.9970 (SELF-ENSEMBLE).

Method	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$	Q2n $\uparrow$	CC $\uparrow$	SCC $\uparrow$	RMSE (DN) $\downarrow$
GSA [4]	47.5769	0.9933	1.7216	2.1667	0.9658	0.9894	0.9773	70.54
CNMF [5]	44.5291	0.9891	1.7259	2.8676	0.9662	0.9828	0.9645	101.64
FUSE [7]	41.1550	0.9787	2.5814	3.4710	0.9418	0.9736	0.9486	148.10
U2Net [36]	50.5455	0.9956	1.2389	1.8740	0.9928	0.9920	0.9798	49.97
PSRT [10]	50.6722	0.9952	1.2733	2.0107	0.9862	0.9906	0.9778	49.00
DCT [44]	52.1998	0.9977	0.9239	1.7448	0.9832	0.9935	0.9873	41.45
DDIF [14]	53.4421	0.9972	0.9387	1.8087	0.9963	0.9929	0.9840	35.58
MIMFormer [28]	53.3578	0.9973	0.9094	1.7611	0.9948	0.9931	0.9850	35.90
MIMO-SST [11]	53.6073	0.9972	0.9315	1.8119	0.9944	0.9926	0.9834	34.93
OTPNNet [37]	52.8243	0.9969	1.0201	1.9332	0.9940	0.9916	0.9841	38.49
SMGU [41]	54.5244	0.9977	0.8719	1.6405	0.9952	0.9939	0.9875	31.29
AMSF [29]	49.3827	0.9944	1.4852	2.3943	0.9878	0.9879	0.9790	56.78
SSDT [13]	52.4788	0.9975	0.8768	1.7992	0.9939	0.9928	0.9847	40.00
ASMNet [30]	53.4528	0.9972	0.9472	1.9153	0.9874	0.9917	0.9848	35.89
DPFormer [31]	53.1632	0.9970	0.9483	1.8795	0.9949	0.9923	0.9824	37.18
CLSNet [43]	55.2007	0.9979	0.7836	1.6318	0.9943	0.9941	0.9869	29.06
Two-stage diffusion [15]	56.1871	0.9982	0.7203	1.5158	0.9969	0.9948	0.9891	25.73
PUFormer (single pass)	58.1467	0.9989	0.7171	1.3367	0.9930	0.9967	0.9992	19.06
PUFormer (self-ensemble)	58.2016	0.9989	0.7126	1.3304	0.9931	0.9967	0.9992	18.94

TABLE IV

METHODS RUN UNDER OUR PIPELINE, WITH ALL PSNR AND SSIM DEFINITIONS. P_g : PEAK = DATASET MAXIMUM (HEADLINE); P_i : PEAK = IMAGE MAXIMUM; P_b : BAND-WISE MEAN, PEAK 1. S_1 : SSIM, DATA RANGE 1; S_i : DATA RANGE = IMAGE MAXIMUM.

Method	P_g	P_i	P_b	S_1	S_i	SAM	RMSE
<i>Chikusei</i>							
Bicubic	33.63	28.20	35.25	0.8500	0.7404	4.406	321.8
GSA [4]	43.20	37.76	44.82	0.9859	0.9778	2.730	107.7
SSR-NET [8]	49.98	44.55	53.83	—	0.9922	1.761	48.8
PUFormer, first run	57.99	52.56	65.29	0.9989	0.9969	0.728	19.3
PUFormer v3	58.15	52.71	66.26	0.9989	0.9969	0.717	19.1
PUFormer v3, ensemble	58.20	52.77	66.33	0.9989	0.9970	0.713	18.9
<i>Pavia Centre</i>							
Bicubic	25.52	25.11	25.87	0.6427	0.6282	6.555	436.3
GSA [4]	34.78	34.37	35.19	0.9450	0.9438	5.229	157.9
PUFormer	50.63	50.22	51.11	0.9974	0.9970	1.895	23.7
PUFormer, ensemble	50.69	50.28	51.17	0.9974	0.9970	1.883	23.5

analyzes. Because the RMSE is expressed in sensor digital numbers, it does not depend on any choice of peak, and it gives the most direct comparison across papers. Table IV lists our own runs of reference methods under the identical pipeline and all PSNR and SSIM variants. Our GSA implementation scores 4.4 dB below the value reported in [15], so our protocol does not favor classical methods.

B. Comparison with the state of the art on Pavia Centre

Table V compares PUFormer with the 17 methods of [15] on Pavia Centre. Trained from scratch in one session, PUFormer reaches 50.69 dB PSNR, 3.53 dB above the best reported value, and lowers the RMSE by 34.1% and ERGAS by 25.3%. It ranks first on six of the eight metrics (PSNR, SSIM in the PSRT definition, ERGAS, CC, SCC and RMSE), but 13th on SAM (1.883° against 1.4542°) and 16th on Q2n (0.9753 against 0.9987). With the stricter per-image data range its SSIM is 0.9970, below the reported 0.9972. The PSNR gain is larger than on Chikusei although the training region is

eight times smaller (97 against 800 non-overlapping patch equivalents).

Both shortfalls come from one test image (Table VI, Fig. 2). On images 1, 3 and 4, SAM is 1.55–1.68° and Q2n 0.9987–0.9991, at or above the best reported Q2n. Image 2 contains a stretch of the river Ticino: 63% of its pixels have a positive normalized difference water index (green against near infrared), against 11, 7 and 13% in the other images, and its SAM is 2.72° and its Q2n 0.904. Water reflects little light, so its spectra are dominated by sensor noise that neither observation carries. The per-band noise that we estimate from the ground truth alone makes an angle of 3.8° with the noise-free spectrum in the darkest tenth of the pixels, against 0.23° in the brightest tenth. Over water PUFormer's SAM is 2.95°, over all other pixels 1.56°, and the darkest 10% of the pixels account for 34% of its mean SAM. The same pixels make image 2 nearly flat within many 32×32 blocks, where Q2n divides the error by a very small local variance; image 2 alone carries 97% of the Q2n deficit $\sum_i (1 - Q2n_i)$ of the four images. Whether the test images of [15] contain water is not stated, so on this dataset SAM and Q2n are the least comparable of the eight metrics and RMSE the most.

As on Chikusei, our GSA scores below the value reported for it (34.78 against 35.35 dB), so the rebuilt data do not favor our method. The reconstruction reproduces its inputs to within 79.9 dB (LR-HSI) and 77.6 dB (HR-MSI). Validation PSNR stops improving after 43,000 iterations (Fig. 3), and those weights are tested; the test PSNR, logged but never used, peaks at 50.66 dB at 25,000 iterations, against 50.63 dB for the selected weights.

C. Per-image results and visual quality

Table VI gives the results of every test image. Chikusei images 5–8 are darker (their maximum is 0.36–0.47 of the dataset maximum), which lowers their strict SSIM but not their RMSE. Fig. 4 shows Chikusei image 7, the hardest test

TABLE V

PAVIA CENTRE $\times 4$ COMPARISON. ALL ROWS EXCEPT PUFORMER ARE REPORTED IN [15] ON THE BENCHMARK DATA RELEASED WITH PSRT [10]; PUFORMER IS EVALUATED ON OUR SIMULATION OF THE SAME PROTOCOL FROM THE ORIGINAL SCENE (SECTION V). BEST VALUE PER COLUMN IN BOLD. SSIM OF PUFORMER USES THE PSRT DEFINITION (DATA RANGE 1); WITH DATA RANGE EQUAL TO EACH IMAGE'S MAXIMUM IT IS 0.9970 (SINGLE PASS) AND 0.9970 (SELF-ENSEMBLE).

Method	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$	Q2n $\uparrow$	CC $\uparrow$	SCC $\uparrow$	RMSE (DN) $\downarrow$
GSA [4]	35.3495	0.9736	5.0812	2.7260	0.9818	0.9832	0.9768	137.78
CNMF [5]	32.1693	0.9469	3.8494	3.8446	0.9489	0.9683	0.9294	198.26
FUSE [7]	34.7418	0.9612	5.5628	3.2042	0.9103	0.9779	0.9469	146.90
U2Net [36]	44.3439	0.9963	1.6542	1.0924	0.9964	0.9973	0.9963	49.10
PSRT [10]	45.8875	0.9970	1.5230	0.9750	0.9974	0.9978	0.9966	41.85
DCT [44]	46.6745	0.9971	1.5169	0.8753	0.9973	0.9983	0.9972	37.66
DDIF [14]	46.2530	0.9970	1.5253	0.9125	0.9960	0.9982	0.9970	39.56
MIMFormer [28]	44.8497	0.9959	1.8225	1.0683	0.9957	0.9975	0.9954	46.19
MIMO-SST [11]	46.6334	0.9969	1.5587	0.8607	0.9975	0.9984	0.9972	37.68
OTPNNet [37]	45.8081	0.9967	1.6573	0.9465	0.9987	0.9980	0.9968	41.51
SMGU [41]	47.1239	0.9972	1.4690	0.8547	0.9981	0.9984	0.9974	37.09
AMSF [29]	43.4465	0.9950	2.0533	1.2728	0.9929	0.9965	0.9940	54.25
SSDT [13]	46.3173	0.9965	1.6439	0.8918	0.9968	0.9982	0.9969	39.01
ASMNNet [30]	46.2072	0.9966	1.6517	0.9019	0.9940	0.9982	0.9969	39.50
DPFormer [31]	42.4280	0.9948	1.9367	1.3986	0.9945	0.9957	0.9917	60.96
CLSNet [43]	46.5277	0.9969	1.5708	0.8412	0.9983	0.9983	0.9975	36.21
Two-stage diffusion [15]	47.1597	0.9972	1.4542	0.8262	0.9980	0.9985	0.9976	35.71
PUFormer (single pass)	50.6299	0.9974	1.8955	0.6206	0.9751	0.9994	0.9995	23.68
PUFormer (self-ensemble)	50.6915	0.9974	1.8833	0.6170	0.9753	0.9994	0.9995	23.52

TABLE VI

PER-IMAGE RESULTS OF THE FINAL MODELS (SELF-ENSEMBLE)

Image	PSNR	S_i	S_1	SAM	ERGAS	Q2n
<i>Chikusei, PUFormer v3</i>						
1	59.15	0.9994	0.9994	0.644	1.550	0.9957
2	60.45	0.9985	0.9994	0.643	0.972	0.9967
3	60.97	0.9987	0.9995	0.597	1.093	0.9962
4	57.10	0.9981	0.9988	0.862	1.334	0.9917
5	57.29	0.9951	0.9986	0.726	1.304	0.9929
6	57.51	0.9957	0.9987	0.669	1.357	0.9913
7	56.24	0.9953	0.9984	0.830	1.449	0.9913
8	56.90	0.9947	0.9986	0.729	1.584	0.9889
Mean	58.20	0.9970	0.9989	0.713	1.330	0.9931
<i>Pavia Centre, PUFormer</i>						
1	50.48	0.9977	0.9977	1.546	0.555	0.9987
2	52.30	0.9949	0.9964	2.722	0.813	0.9044
3	49.53	0.9979	0.9979	1.677	0.557	0.9991
4	50.45	0.9975	0.9976	1.588	0.543	0.9991
Mean	50.69	0.9970	0.9974	1.883	0.617	0.9753

image, and Fig. 5 Pavia Centre image 3, the hardest there. The errors of bicubic interpolation and GSA follow every field boundary and road (mean absolute errors of 164.7 and 64.8 DN), whereas both PUFormer models keep the error small and nearly uniform over the image (11.5 and 11.0 DN), apart from a few isolated pixels.

Fig. 6 shows the error of every band. PUFormer is more accurate than GSA in all 128 bands, and the final model improves on the first run in every band, most at the red edge (700–760 nm), where the mean band RMSE falls from 7.9 to 4.2 DN (from 9.8 to 1.1 DN at 724 nm). The error is largest in the first eight bands (363–404 nm), which lie below the first WorldView-2 band and therefore receive no high-resolution detail from the multispectral image, and in the last bands near 1000 nm, where the sensor's signal-to-noise ratio drops.

TABLE VII

FROM THE FIRST RUN TO THE FINAL MODEL (TEST SET UNLESS MARKED). CONS.: PSNR($\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_H$) / PSNR($\mathcal{R}\hat{\mathbf{X}}, \mathbf{Y}_M$) IN dB

Model	Val.	PSNR	SAM	ERGAS	Cons.
GSA	–	43.20	2.730	3.279	45.4 / 45.0
First run (zero border)	58.60	57.99	0.728	1.356	76.7 / 75.0
+ self-ensemble	–	58.02	0.726	1.353	76.8 / 75.0
First-run weights, exact border	49.52	54.24	0.773	1.474	–
v3 (exact border, full loss)	58.72	58.15	0.717	1.337	85.5 / 89.0
+ self-ensemble	–	58.20	0.713	1.330	86.0 / 89.5

D. Training stages and the exact border (Chikusei)

Table VII traces the path from the first run to the final model. Evaluating the first-run weights with the exact reflect-border operator, without retraining, drops the validation PSNR from 58.60 to 49.52 dB. The prior had learned to undo the error that the zero-padded operator introduces at the border, so with the exact operator it over-corrects there; the effect is strongest on the small 64×64 validation patches, where border pixels are a large fraction of the image. Fine-tuning removes this within 2000 iterations (58.60 dB validation) and ends at 58.72 dB. The consistency of the reconstruction with its inputs rises from 76.7 to 85.5 dB for the LR-HSI and from 75.0 to 89.0 dB for the HR-MSI. Fig. 7 shows the validation curves; validation PSNR stops improving after about 44,000 fine-tuning iterations, and the test PSNR stays between 58.13 and 58.15 dB from iteration 26,000 onward.

E. Robustness and test-time operator swap

Published methods are usually trained and tested with the same degradation. We test the final model on inputs made with a different blur or a shifted spectral response, and with Gaussian noise added to both observations (Table VIII, Fig. 8). Because the operators are explicit, PUFormer can also be given

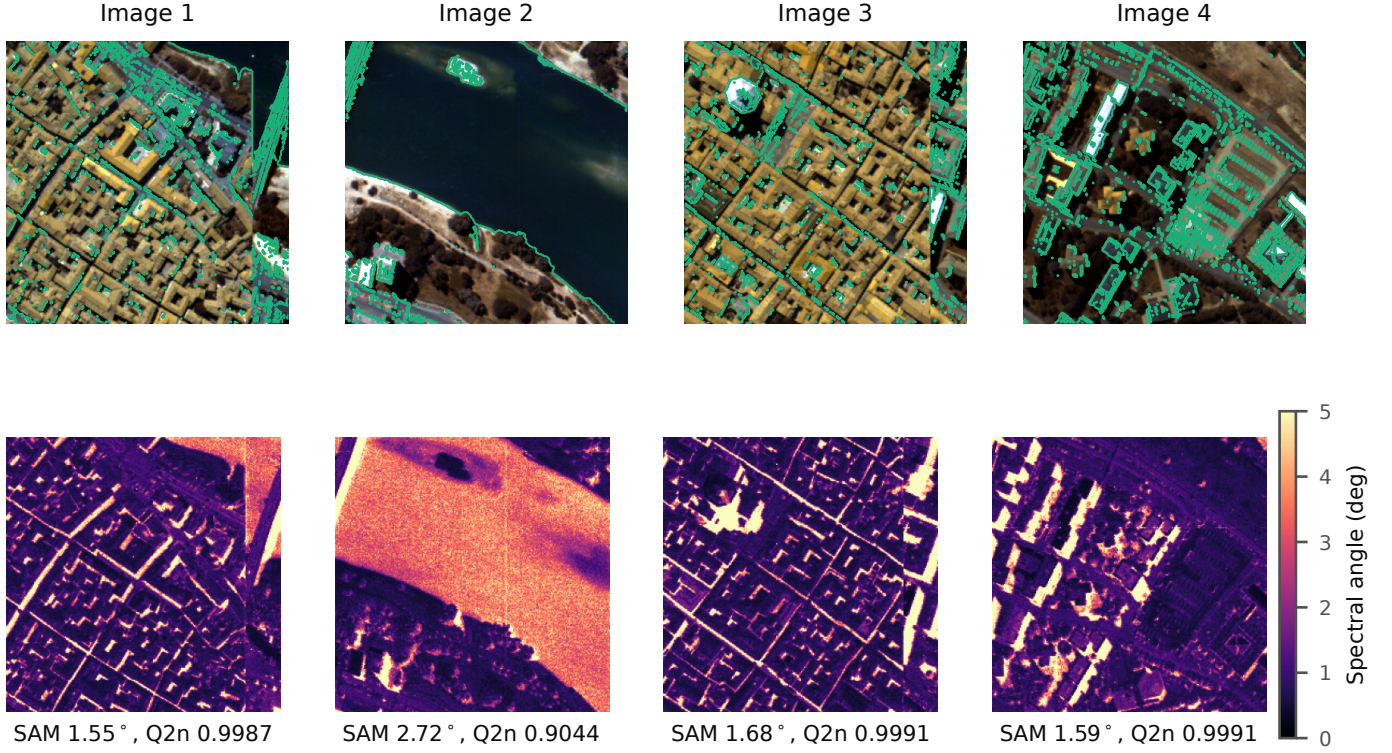


Fig. 2. The four Pavia Centre test images in false color (bands 57, 34 and 3; top), with water pixels (NDWI > 0) outlined, and the spectral angle of PUFormer's reconstruction at every pixel (bottom; color scale clipped at 5°).

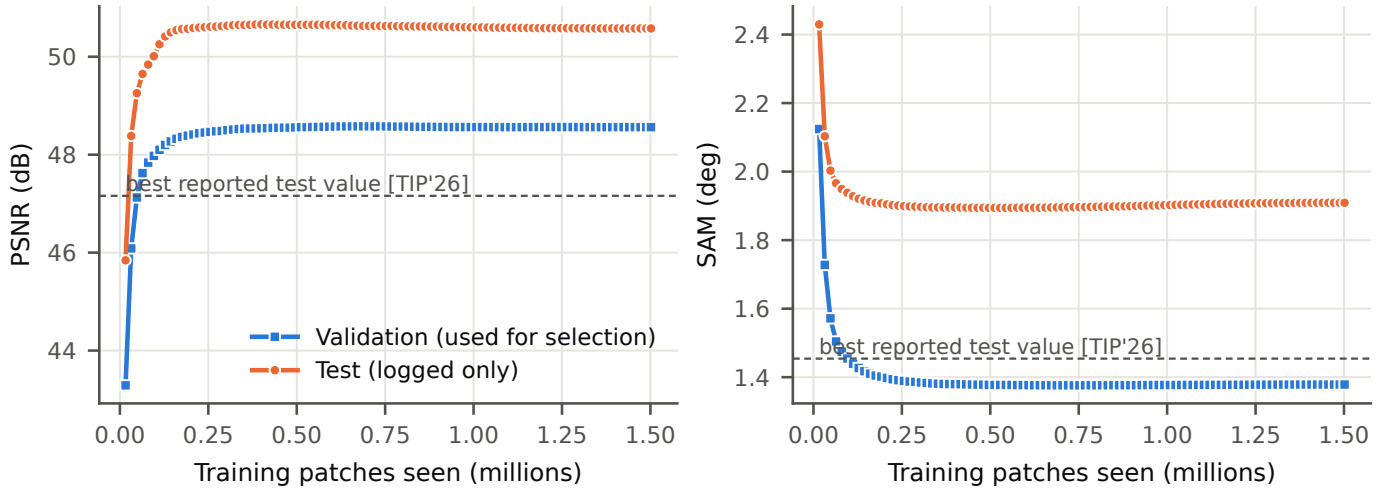


Fig. 3. Pavia Centre: PSNR and SAM during training from scratch, on the validation patches (used to select the weights) and on the test images (logged only). The dashed line is the best value reported in [15] on the PSRT benchmark data.

the true test-time operator without retraining (“swap”). With the training operator, a blur mismatch costs 3.9–7.5 dB. With the true blur swapped in, the loss is at most 0.26 dB for σ between 1.5 and 3.0; with the zero-border operators of the first run, the same swap recovered only to 55.2–57.1 dB. A shifted spectral response is not compensated by the swap, which lowers the PSNR by a further 0.5 dB: the priors, trained with one response, rely on it beyond the data step. Noise is the main weakness of a model trained on noise-free data; PUFormer still exceeds GSA by 10.0 dB at 40 dB SNR and by 2.8 dB at 30 dB SNR. On Pavia Centre the blur behaves the same way:

the training operator loses 3.3–6.8 dB and the swapped-in true operator at most 0.11 dB. Unlike on Chikusei, swapping in a shifted IKONOS response raises the PSNR slightly (by 0.2–0.3 dB), perhaps because an 8 nm shift is small relative to the 66–96 nm IKONOS bands, and the margins over GSA under noise are larger (12.8, 9.8 and 6.0 dB at 40, 35 and 30 dB SNR).

F. Where the Q2n gap on Chikusei comes from

Q2n normalizes every band of every 32×32 block by the ground truth's own mean and standard deviation in that block,

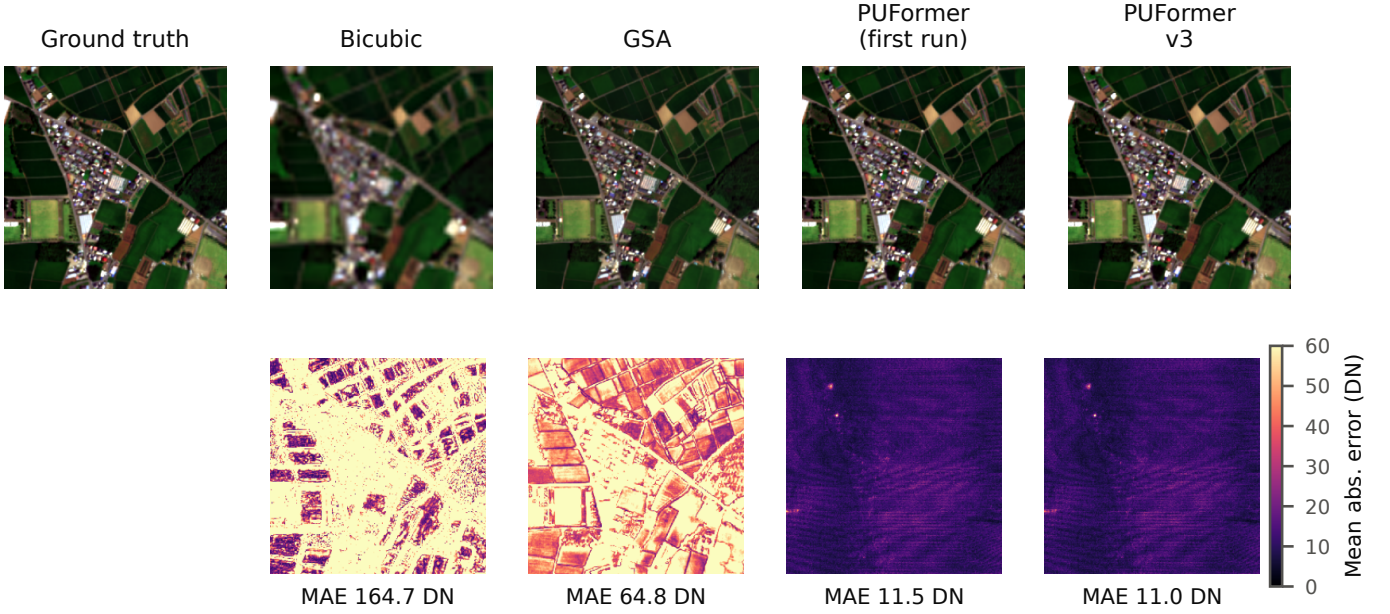


Fig. 4. Chikusei test image 7 in false color (bands 60, 40 and 20; top) and the mean absolute error over all bands in DN (bottom; color scale clipped at 60 DN).

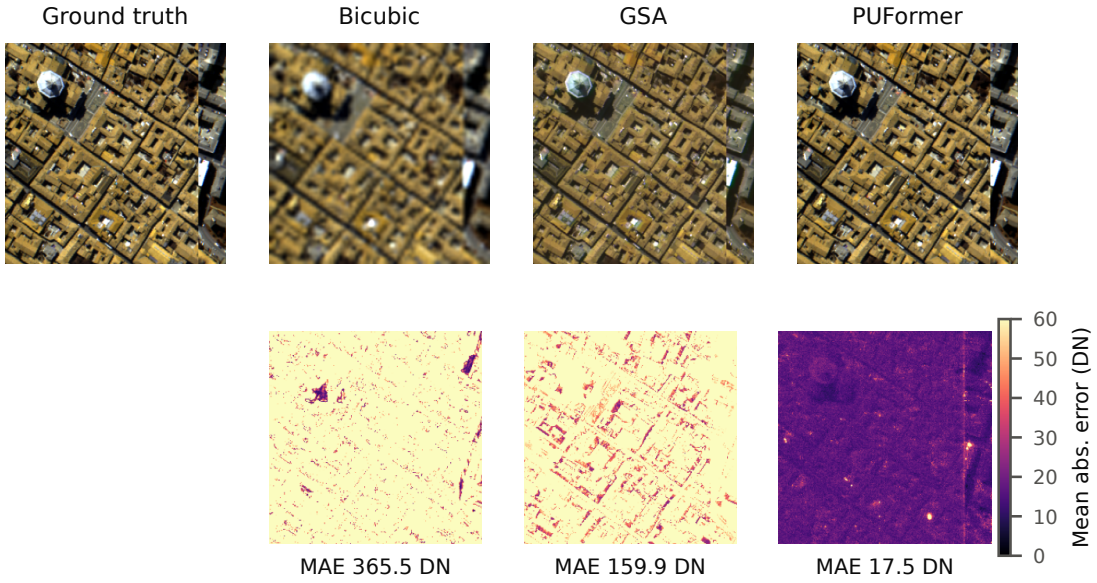


Fig. 5. Pavia Centre test image 3 in false color (bands 57, 34 and 3; top) and the mean absolute error over all bands in DN (bottom; color scale clipped at 60 DN).

so an error that is small in absolute terms counts heavily in a band that is nearly flat within a block. Table IX and Fig. 9 decompose the Q_{2n} of the final model (0.9931). If the eight ultraviolet bands (363–404 nm) were reconstructed exactly, Q_{2n} would rise to 0.9988; if the last 24 bands were exact as well, to 0.9998. In the worst 10% of blocks, the error energy of these eight bands equals 0.41 of the ground truth’s local variance, against at most 0.040 for any other group of eight bands. The fine detail of these bands cannot be predicted from the inputs: a least-squares injection of the multispectral detail, fitted on the test ground truth itself, reduces the band-1 error only from 61.6 to 57.5 DN, while PUFormer reaches 51.8 DN,

and PUFormer is more accurate than this oracle in every band we checked. Q_{2n} is therefore nearly insensitive to overall accuracy on these test images: blending our reconstruction toward bicubic interpolation lowers the PSNR from 58.20 to 52.15 dB but Q_{2n} only from 0.9931 to 0.9904 (Fig. 9, right). The values reported in [15] do not follow PSNR in this way. Between 52.2 and 53.6 dB they range from 0.9832 (DCT) to 0.9963 (DDIF); of the 16 methods within the PSNR range of our curve, 11 lie above it and 5 below. U2Net, for example, is listed at 0.9928 at 50.55 dB, a value our implementation reaches only near 57.6 dB. We conclude that the Q_{2n} values of [15] were most likely computed with a

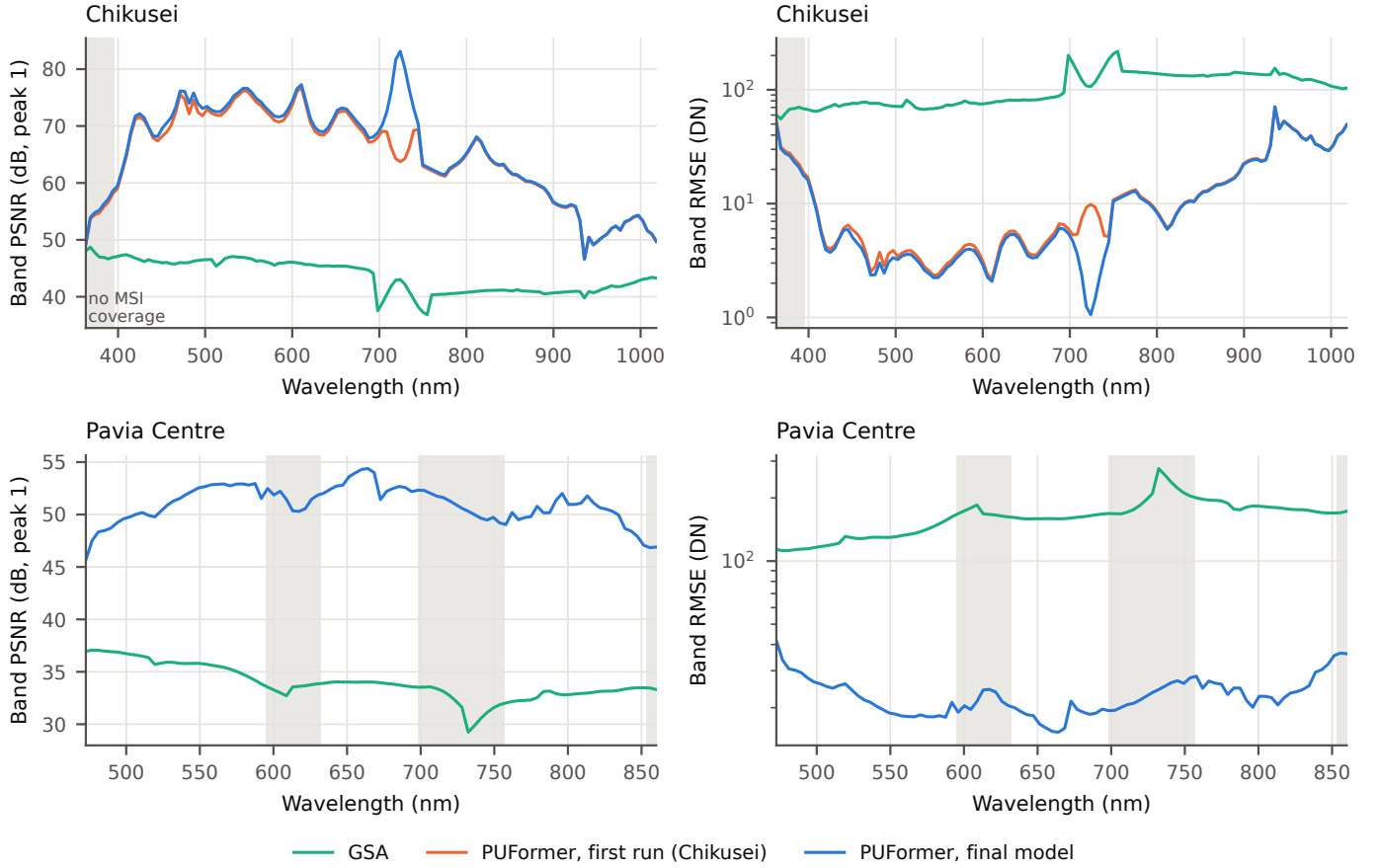


Fig. 6. Band-wise PSNR (left, peak 1) and RMSE in DN (right, log scale) over the test images of Chikusei (top) and Pavia Centre (bottom). Shaded ranges have no multispectral coverage: below the first WorldView-2 band for Chikusei, and between and beyond the IKONOS bands (50% edges) for Pavia Centre.

TABLE VIII

ROBUSTNESS OF THE FINAL MODELS (TEST PSNR in dB / SAM in DEGREES). FIXED: TRAINING OPERATOR; SWAP: TRUE OPERATOR SUPPLIED AT TEST TIME. THE BETTER PUFORMER ENTRY OF EACH CONDITION IS IN BOLD.

Condition	Chikusei (PUFormer v3)			Pavia Centre (PUFormer)		
	GSA	PUFormer fixed	PUFormer swap	GSA	PUFormer fixed	PUFormer swap
Nominal	43.20 / 2.73	58.15 / 0.72	–	34.78 / 5.23	50.63 / 1.90	–
Blur $\sigma = 1.5$	44.05 / 2.57	50.68 / 1.15	57.97 / 0.73	35.59 / 4.99	43.80 / 2.75	50.56 / 1.90
Blur $\sigma = 2.5$	42.77 / 2.82	54.26 / 0.88	58.05 / 0.72	34.36 / 5.36	47.30 / 2.14	50.59 / 1.90
Blur $\sigma = 3.0$	42.54 / 2.87	51.69 / 1.03	57.89 / 0.74	34.13 / 5.43	44.87 / 2.37	50.52 / 1.91
SRF shift -8 nm	42.86 / 2.91	53.01 / 1.20	52.54 / 1.31	34.58 / 5.36	49.46 / 1.99	49.75 / 2.03
SRF shift $+8$ nm	43.29 / 2.68	53.03 / 1.19	52.47 / 1.28	34.79 / 5.20	49.62 / 1.99	49.82 / 2.02
Noise, 40 dB SNR	43.04 / 2.85	53.03 / 1.39	–	34.71 / 5.37	47.55 / 2.89	–
Noise, 35 dB SNR	42.71 / 3.09	49.11 / 2.20	–	34.57 / 5.64	44.36 / 4.19	–
Noise, 30 dB SNR	41.84 / 3.67	44.62 / 3.69	–	34.16 / 6.37	40.18 / 6.60	–

TABLE IX

Q2N ANALYSIS OF PUFORMER V3 (SELF-ENSEMBLE, TEST SET)

Setting	Q2n
PUFormer v3	0.9931
bands 1–8 (363–404 nm) set to ground truth	0.9988
bands 1–8 and 105–128 set to ground truth	0.9998
Deficit share, worst 10% / 50% of blocks	25% / 75%

G. Complexity

PUFormer has 10.13 M parameters and needs 18.6 GFLOPs for a 64×64 patch and 297 GFLOPs for a 256×256 image. On one Tesla T4 it reconstructs a 256×256 image with 128 bands in 0.20 s with mixed precision (92 bands: 0.18 s), and the self-ensemble doubles this. Because the attention is computed across channels, memory and time grow linearly with the number of pixels.

VII. DISCUSSION AND LIMITATIONS

Three caveats apply to the comparison. First, our test images follow the protocols as described in [15], but they may not be

different implementation, normalization or test region, and that they are not directly comparable with ours.

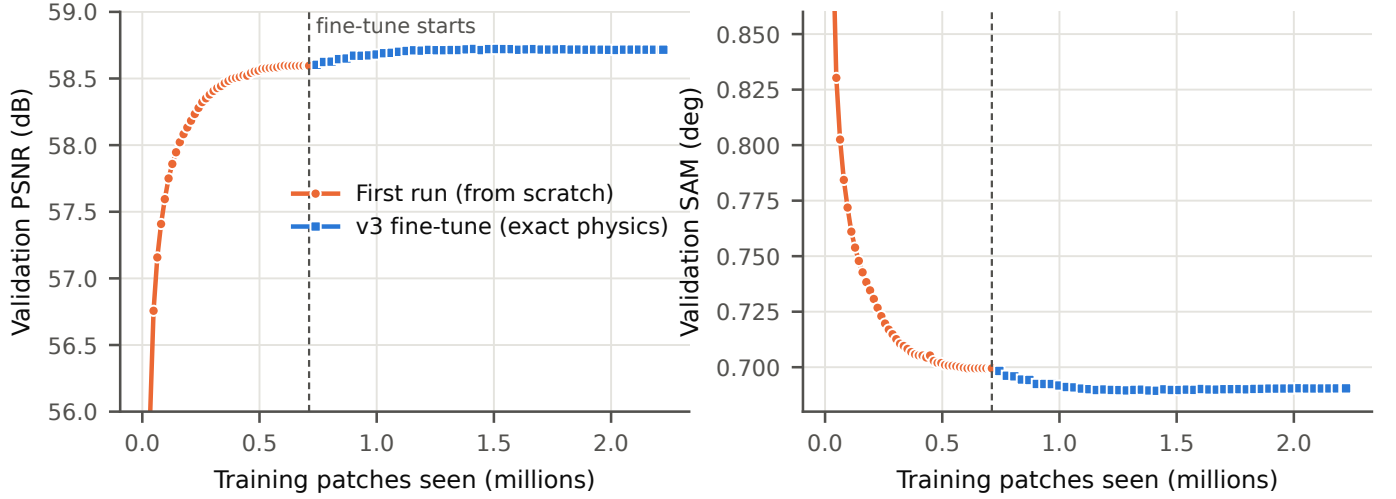


Fig. 7. Validation PSNR and SAM during training, plotted against the number of 64×64 training patches seen. The first run starts from random weights (its first two evaluations lie below the plotted range); v3 continues from its final weights with the exact border and the full loss.

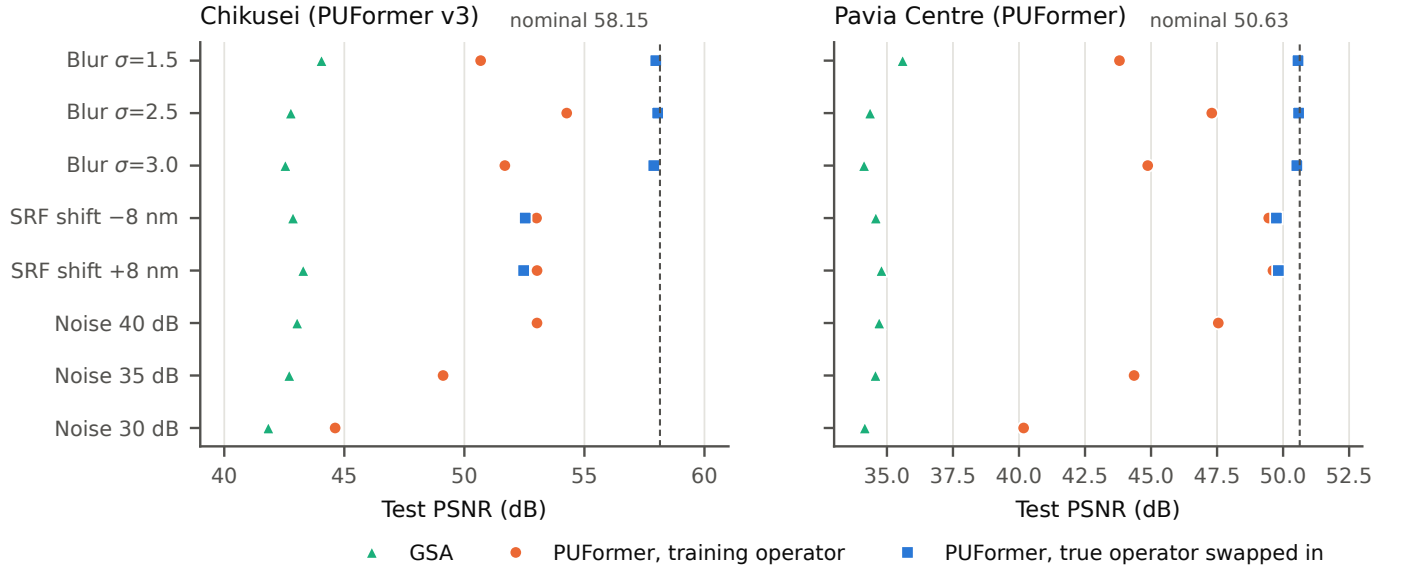


Fig. 8. Test PSNR under degradation mismatch and noise on Chikusei (left) and Pavia Centre (right). The dashed line is the nominal PUFormer result.

the exact pixels used there; for Pavia Centre we also had to rebuild the benchmark data from the original scene, and all other rows of Tables III and V are reported values. RMSE in digital numbers, which does not depend on a peak or data range, is the most robust basis for comparison, and there PUFormer’s lead is largest (18.9 vs. 25.7 DN on Chikusei, 23.5 vs. 35.7 DN on Pavia Centre). Second, SSIM depends strongly on its data range: with the stricter per-image range, PUFormer’s SSIM (0.9970 on both datasets) is below the values reported in [15]. Third, as shown above, Q2n on Chikusei is dominated by bands that no method can reconstruct well, and SAM and Q2n on Pavia Centre by one test image with water. Beyond the comparison, the model is trained without noise and for one sensor pair; the swap experiment shows that the exact operators make the network adaptable to the blur but not to the spectral response, and training with noise and with randomized responses is a natural next step.

Evaluation on real, not simulated, image pairs is left for future work.

VIII. CONCLUSION

PUFormer combines exact data-consistency steps, including the correct image border and its adjoint, with a transposed-attention prior and cross-stage memory. Under a fully specified Chikusei $\times 4$ protocol it improves on the best reported result by 2.0 dB PSNR and 26% RMSE and ranks first on seven of eight metrics, while its reconstructions reproduce both observations to within 86–90 dB. On Pavia Centre, trained from scratch in one session, it improves on the best reported result by 3.5 dB PSNR and 34% RMSE and ranks first on six of eight metrics; its SAM and Q2n trail because of one test image that contains water. The exact operators also let the model absorb a blur mismatch at test time with almost no loss. All code, training

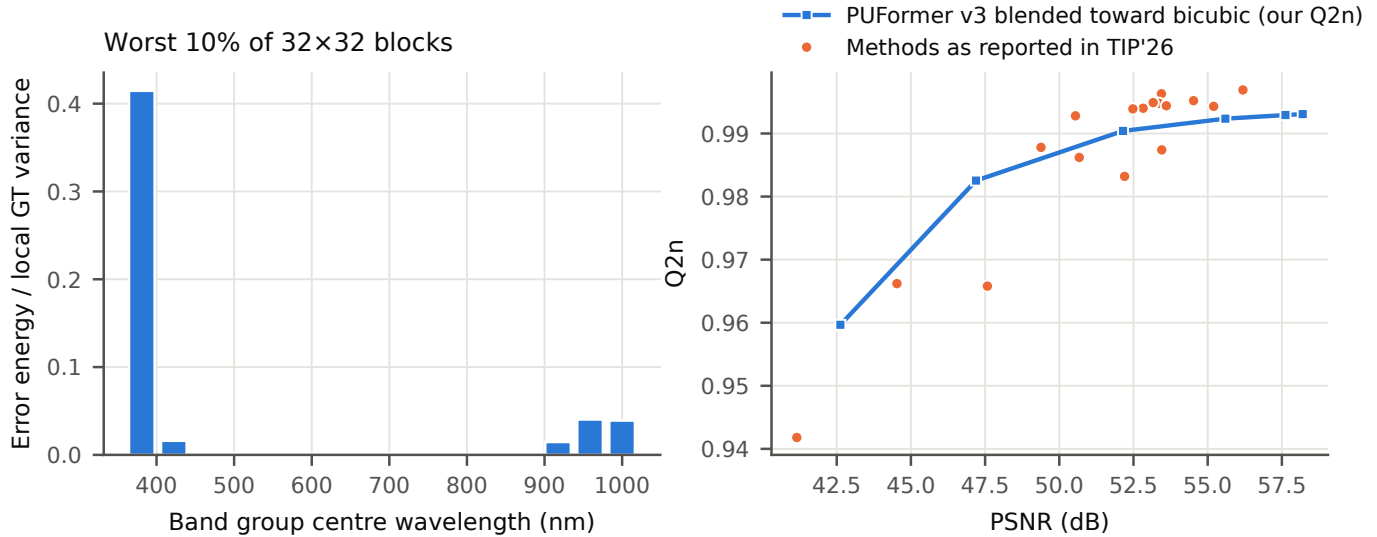


Fig. 9. Left: error energy relative to the local ground-truth variance per group of eight bands in the worst 10% of 32×32 blocks. Right: Q2n against PSNR when PUFormer's output is blended toward bicubic interpolation (our implementation), and the (PSNR, Q2n) pairs reported in [15].

logs, result files and the evaluation scripts, including the metric implementations, are publicly available.

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